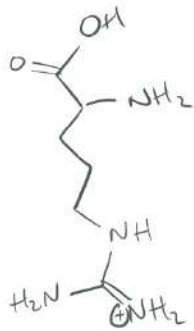
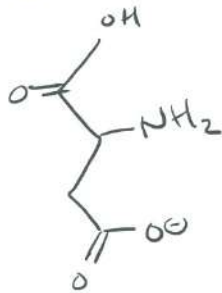


Arginine
Arg / R
Basic



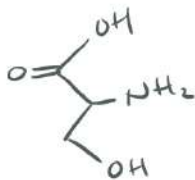
*guanidino

Aspartic Acid
Asp / D
Acidic



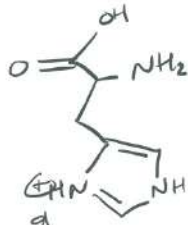
* 2 carbon chain

Serine
Ser / S
Polar



* OH on last carbon

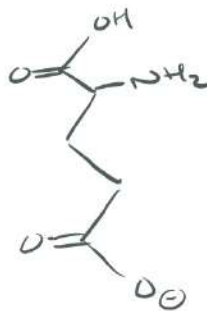
Histidine
His / H
Basic



most acidic proton (resonance)

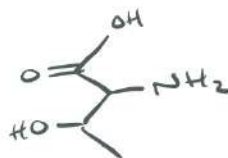
* imidazole group

Glutamic Acid
Glu / E
Acidic



* 3 carbon chain

Threonine
Thr / T
Polar

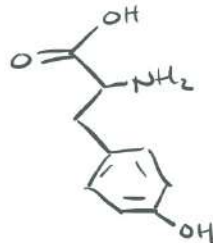


* OH on 2nd to last carbon

Lysine
Lys / K
Basic



Tyrosine
Tyr / Y
Polar



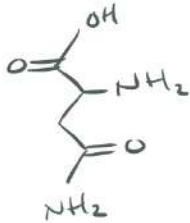
* aromatic

targets of phosphorylation

Asparagine

Asn / N

Polar

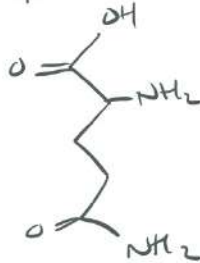


* 2 carbon chain

Glutamine

Gln / Q

Polar

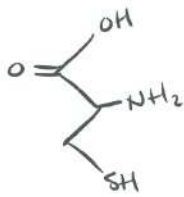


* 3 carbon chain

Cysteine

Cys / C

Nonpolar

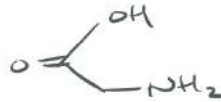


* only amino acid that can form disulfide bonds

Glycine

Gly / G

Nonpolar



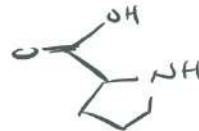
* only achiral amino acid

* disrupts alpha-helices due to excessive flexibility

Proline

Pro / P

Nonpolar



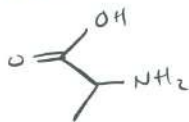
* found @ turns of β -pleated sheets

* can interrupt 2^o protein structure due to structural rigidity; least likely to be found in alpha-helices

Alanine

Ala / A

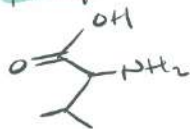
Nonpolar



Valine

Val / V

Nonpolar

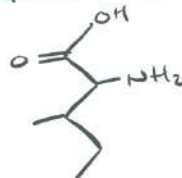


* 2 carbon side chain

Isoleucine

Ile / I

Nonpolar



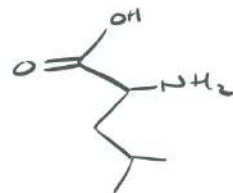
* Methyl on first carbon of side chain

* 3 carbon side chain

Leucine

Leu / L

Nonpolar



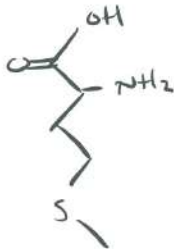
* Methyl group on second carbon of side chain

* 3 carbon side chain

Methionine

Met / M

Nonpolar

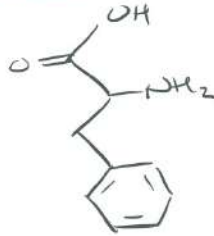


* does NOT create disulfide bridge

Phenylalanine

Phe / F

Nonpolar

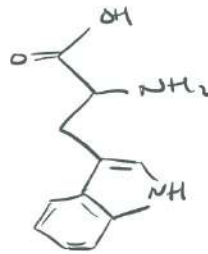


* aromatic

Tryptophan

Trp / W

Nonpolar



* aromatic

* indole group

Important things to note:

- Alcohol-containing amino acids:
 - Serine
 - Threonine
 - Tyrosine
- Stereochemistry is S for all except cysteine
→ Digestible in L configuration
- Avg. weight of a.a. = 110 Daltons
- Tryptophan is the largest amino acid
- Side chain pKa values: DEHCYKR
 - D (aspartate) ~ 4
 - E (glutamate) ~ 4
 - H (histidine) ~ 6
 - C (cysteine) ~ 8
 - Y (tyrosine) ~ 10
 - K (lysine) ~ 11
 - R (arginine) ~ 12

• Ketogenic vs. glucogenic

↓ fatty acid synthesis	↓ glucose synthesis
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- Ketogenic: Leucine + Lysine (the "L's")
- Glucogenic: The remaining 18
- Glucogenic AND Ketogenic: Phenylalanine, Isoleucine, Tryptophan, Tyrosine, Threonine ("FITT")